

ZHIYANG CHEN

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EDUCATION

University of Toronto

Ph.D. in Computer Science, supervised by Prof. Fan Long	May 2023 – 2027 (expected)
Visiting PhD, ETH Zurich (ETH AI Center), hosted by Prof. Zhendong Su	May 2026 – Jan. 2027
M.Sc. in Computer Science, supervised by Prof. Fan Long	Sept. 2021 – Feb. 2023

University of Michigan, Ann Arbor

B.S.E. in Computer Science Engineering, Minor in Mathematics, Summa Cum Laude	Sept. 2019 – May 2021
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Shanghai Jiao Tong University

B.S. in Electrical and Computer Engineering, Minor in Entrepreneurship	Sept. 2017 – Aug. 2021
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SELECTED PUBLICATIONS

Zhiyang Chen, Fan Long, and Zhendong Su. **Ghost-Filled Orders: Detecting and Testing Atomicity Violations in Non-Custodial Prediction Markets**. *arXiv preprint arXiv:2609.17902*, 2026.

Tara Saba, **Zhiyang Chen**, Jikai Jason Li, Anne Ouyang, Xujie Si, and Fan Long. **CuTeGen: LLM-Driven Agentic Generation of High-Performance GPU Kernels in CuTe**. *Under submission*.

Zhiyang Chen, Tara Saba, Xun Deng, Xujie Si, and Fan Long. **Scam2Prompt: A Scalable Framework for Auditing Malicious Scam Endpoints in Production LLMs**. *International Conference on Machine Learning (ICML)*, 2026.

Zhiyang Chen, Sidi Mohamed Beillahi, Pasha Barahimi, Cyrus Minwalla, Han Du, Andreas Veneris, and Fan Long. **Enforcing Control Flow Integrity on DeFi Smart Contracts**. *IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026. Project funded by and conducted in collaboration with the **Bank of Canada**.

Zhiyang Chen, Ye Liu, Sidi Mohamed Beillahi, Yi Li, and Fan Long. **OpenTracer: A Dynamic Transaction Trace Analyzer for Smart Contract Invariant Generation and Beyond**. *IEEE/ACM International Conference on Automated Software Engineering (ASE), Tool Demonstrations Track*, 2024.

Zhiyang Chen, Ye Liu, Sidi Mohamed Beillahi, Yi Li, and Fan Long. **Demystifying Invariant Effectiveness for Securing Smart Contracts**. *ACM International Conference on the Foundations of Software Engineering (FSE)*, 2024.

Zhiyang Chen, Sidi Mohamed Beillahi, and Fan Long. **FlashSyn: Flash Loan Attack Synthesis via Counterexample-Driven Approximation**. *IEEE/ACM International Conference on Software Engineering (ICSE)*, 2024.

Tianyi Zhang, **Zhiyang Chen**, Yuanli Zhu, Priyan Vaithilingam, Xinyu Wang, and Elena L. Glassman. **Interpretable Program Synthesis**. *ACM Conference on Human Factors in Computing Systems (CHI)*, 2021.

AWARDS AND DISTINCTIONS

Scholarships and Grants

Canada Graduate Research Scholarship – Doctoral (3 years)	CAD 40,000/year	2026
ETH Zurich Academic Guest	CHF 4,500	2026
Bell Graduate Scholarship, University of Toronto	CAD 20,000	2024
Mitacs Accelerate Fellowship, Bank of Canada / U of T	CAD 20,000	2023–2024
Co-author, Ethereum Foundation Grant FY23-0882		2023
Smart Contract Research Forum Grant	USD 1,000	2022
Wolfond Scholarship in Wireless Information Technology (U of T)	CAD 10,000	2021

Honors and Professional Recognition

SQLite Committer (see my commits) — focused on core database performance optimization	2026
Outstanding Graduate, Shanghai Jiao Tong University	2021
James B. Angell Scholar, University of Michigan	2021
University Honours and Dean’s List every term, University of Michigan	2019–2021
Elected Member, Tau Beta Pi–Michigan Gamma engineering honor society	Nov. 2020

RESEARCH IMPACT

Scam2Prompt (ICML 2026). Research disclosures led to **62 previously unlisted phishing websites** being confirmed and added to MetaMask’s phishing-detection database.

Sequencer Level Security (ICBC 2025 Poster). Co-authored research deployed on Zircuit mainnet to screen transactions before block inclusion and quarantine suspected exploits. Media: *The Block*.

Exploit Replay and EVM Instrumentation (2025). Cited in **Zircuit’s 2025 roadmap**: WETSEB research enabled end-to-end SLS testing; IWBOSE research informed planned invariant checks in SLS 2.0/3.0.

OpenTracer (ASE 2024). The **first open-source EVM transaction visualizer** combining decoded function calls with storage-state changes. Powered Trace2Inv analysis of **350,800 transactions** across **23 invariant templates** in eight categories.

FlashSyn (ICSE 2024). Adopted by Quantstamp as **Economic Exploit Analysis** to detect flash-loan vulnerabilities in deployed and pre-deployment EVM protocols. Media: *CoinDesk*, *The Block*.

PROFESSIONAL EXPERIENCE

Research Consultant, Zircuit May 2024 – Jan 2026

- Co-authored *Sequencer Level Security* for Zircuit, an Ethereum Layer 2 blockchain with over USD 2 billion in ecosystem TVL (2025); this signature launch feature screened transactions before block inclusion and quarantined suspected exploits.
- Developed research on historical exploit replication and EVM trace instrumentation, cited in Zircuit’s 2025 technical roadmap as foundations for testing and extending SLS.

Research Consultant, Quantstamp May 2023 – May 2024

- Helped translate **FlashSyn** research into **Economic Exploit Analysis**, Quantstamp’s production service for detecting flash-loan vulnerabilities in EVM protocols.
- Extended FlashSyn’s synthesis engine and conducted static analysis of smart-contract logic for automated vulnerability detection.

Research Intern, Quantstamp Feb 2023 – May 2023

- Optimized the synthesis algorithm of FlashSyn by pruning the search space, which improved tool scalability and reduced execution time for complex benchmarks.

SERVICE

Program Committee: ICML 2026

Journal Reviewer: IEEE TON (2025), IEEE TSE (2024–2025), ACM TOSEM (2025–2026), IEEE TNSM (2024–2025), and AUSE (2025)

Artifact Evaluation Committee: SOSP (2025), CAV (2025), ICSE (2025), and PLDI (2024)

Shadow/Junior Program Committee: ICSE (2026) and MSR (2025)

Sub-reviewer: ICSE (2027), POPL (2027), ICSE (2024), ASE (2024), and FSE (2023)

Session Chair, Sponsor Industry Forum, ICFP/SPLASH (2025); Hackathon Judge, ETHDenver (2025)

TEACHING EXPERIENCE

University of Toronto (Teaching Assistant) Sep 2021 – Dec 2025

- **CSC2125: Topics in Software Engineering.** Fall 2025
- **CSC488/2107: Compilers and Interpreters (Head TA).** Winter 2023
- **CSC373: Algorithm Design, Analysis & Complexity.** Fall 2022
- **CSC263: Data Structures and Analysis.** Winter 2022
- **CSC108: Introduction to Computer Programming.** Fall 2021

University of Michigan (Grader) Fall 2020 – Winter 2021

- **MATH 450: Advanced Mathematics for Engineers I.** Winter 2021
- **MATH 214: Applied Linear Algebra.** Fall 2020